

Software Requirements Specification (SRS)

ERP System for Academic, Marketing & Finance, and Administration & HR

Course Title: Large System
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Table of Contents

Section	Page
1. Introduction	4
1.1 Purpose	4
1.2 Document Conventions	4
1.3 Intended Audience	4
1.4 Product Scope	5
1.5 References	5
2. Overall Description	5
2.1 Product Perspective	5
2.2 Product Functions	5
2.3 User Classes and Characteristics	6
2.4 Operating Environment	6
2.5 Design and Implementation Constraints	7
3. External Interface Requirements	7
3.1 User Interfaces	7
3.2 Hardware Interfaces	7
3.3 Software Interfaces	8

Section	Page
3.4 Communications Interfaces	8
4. System Features	8
4.1 User Authentication System	8
4.2 Database Management	9
4.3 API Integration	9
4.4 Academic Module	9
4.5 Marketing & Finance Module	10
4.6 Administration & Human Resources Module	11
5. Nonfunctional Requirements	12
5.1 Performance Requirements	12
5.2 Safety Requirements	12
5.3 Security Requirements	13
5.4 Software Quality Attributes	13
5.5 Business Rules	13
6. System Models	14
6.1 Use Case Diagrams	14
6.2 Class Diagram	16
6.3 Sequence Diagrams	17
Appendix C: Additional Details	20
References	22

1. Introduction

1.1 Purpose

This SRS document specifies the requirements for an Enterprise Resource Planning (ERP) system designed for educational institutions, integrating Academic Management, Marketing & Finance, and Administration & Human Resources modules.

1.2 Document Conventions

- Bold text:** Important terms or emphasis
- Italic text:* System components or module names
- `Code formatting:` Technical elements, API endpoints, database fields
- Priority Levels:** High (Critical), Medium (Important), Low (Nice-to-have)

1.3 Intended Audience

- **Development Team:** Sections 3, 4, and 6
- **Project Managers:** Sections 1, 2, 4, and 5
- **Testing Team:** Sections 4 and 5
- **End Users:** Sections 2 and 4
- **Stakeholders:** Sections 1, 2, and 5

1.4 Product Scope

The ERP system streamlines educational institution operations by integrating academic, financial, marketing, and HR functions into a unified platform.

Goals:

- Reduce manual administrative tasks by 60%
 - Improve data accuracy and consistency
 - Enable real-time access to critical information
 - Enhance user experience through intuitive interfaces
 - Ensure scalability for future expansion
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2. Overall Description

2.1 Product Perspective

A self-contained web-based application that replaces disconnected legacy systems and provides unified data access across departments through:

- Web-based user interface
- RESTful API for inter-module communication
- Database backend (MySQL/PostgreSQL)
- Email server integration
- Payment gateway integration

2.2 Product Functions

- **User Management:** Role-based access control, secure authentication
- **Academic Operations:** Course management, student enrollment, examination management
- **Financial Management:** Fee management, expense tracking, financial reporting

- **Marketing Operations:** Campaign management, lead tracking, ROI analysis
- **Human Resources:** Employee management, payroll, attendance, leave management
- **Asset Management:** Inventory tracking, asset allocation
- **Reporting & Analytics:** Dynamic dashboards, customizable reports

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise	Usage Frequency
System Administrator	System configuration and maintenance	High	Daily
Academic Staff	Course and grade management	Medium	Daily
Students	View courses, grades, make payments	Low-Medium	Daily
Finance Officers	Fee and expense management	Medium	Daily
Marketing Staff	Campaign and lead management	Medium	Daily
HR Managers	Employee and HR operations	Medium	Daily
Employees	Self-service portal access	Low-Medium	Weekly
Management	Reports and analytics	Low-Medium	Weekly

2.4 Operating Environment

Hardware:

- Server: 8GB RAM, 4-core processor, 100GB storage
- Client: Any device with modern web browser

Software:

- Server OS: Linux (Ubuntu 20.04+) or Windows Server 2016+
- Database: MySQL 8.0+ or PostgreSQL 12+
- Web Server: Apache 2.4+ or Nginx 1.18+
- Client Browser: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+

Network:

- Minimum 10 Mbps internet connection
- HTTPS support required

2.5 Design and Implementation Constraints

- Must use relational database (MySQL/PostgreSQL)
- RESTful API architecture required
- Password hashing using bcrypt or SHA-256

- HTTPS for all communications
 - Support minimum 500 concurrent users
 - Page load time under 3 seconds
 - Database normalization to 3NF minimum
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3. External Interface Requirements

3.1 User Interfaces

General Requirements:

- Responsive design (desktop, tablet, mobile)
- Consistent navigation across modules
- WCAG 2.1 Level AA accessibility compliance
- Light/dark mode support

Key Screens:

- Login Screen: Credentials, role selection, password recovery
- Dashboard: Navigation, statistics widgets, notifications
- Data Entry Forms: Validation, required fields, action buttons
- Data Tables: Sortable columns, search, pagination, filters
- Reports: Filters, export options (PDF, Excel, CSV), visualizations

3.2 Hardware Interfaces

- Network connectivity for internet access
- Storage devices for database and backups
- Standard input devices (keyboard, mouse, touchscreen)
- Display screen (minimum 1024×768)
- Optional: Printers, barcode scanners, biometric devices

3.3 Software Interfaces

System	Purpose	Interface	Data Format
MySQL/PostgreSQL	Data storage	SQL via database driver	Prepared statements
SMTP Server	Email notifications	SMTP protocol	HTML/plain text
Payment Gateway	Fee processing	RESTful API	JSON (HTTPS)

System	Purpose	Interface	Data Format
Apache/Nginx	Web serving	HTTP/HTTPS	HTML, CSS, JS

3.4 Communications Interfaces

- **HTTP/HTTPS:** All client-server communication via HTTPS (TLS 1.2+)
- **RESTful API:** JSON format, JWT authentication, standard HTTP methods
- **Database:** TCP/IP with connection pooling
- **Email:** SMTP with TLS encryption

4. System Features

4.1 User Authentication System

Priority: High

Functional Requirements:

REQ-AUTH-01: User Login

- Users log in using username/email and password
- System validates credentials and creates session
- Account locked after 3 failed attempts

REQ-AUTH-02: Password Security

- Passwords hashed using bcrypt or SHA-256
- Minimum 8 characters with uppercase, lowercase, number, special character

REQ-AUTH-03: Role-Based Access Control

- System enforces permissions based on user roles
- Roles: System Administrator, Academic Staff, Student, Finance Officer, Marketing Staff, HR Manager, Employee

REQ-AUTH-04: Session Management

- Secure session tokens with 30-minute inactivity timeout
- Manual logout functionality

REQ-AUTH-05: Password Recovery

- Email-based password reset with expiring tokens

4.2 Database Management

Priority: High

Functional Requirements (REQ-DB-01 to REQ-DB-07):

- MySQL 8.0+ or PostgreSQL 12+ required
- Database normalized to 3NF minimum
- Foreign key constraints enforced
- Automated daily backups (retained 30 days)
- Data restoration capability
- Encrypted connections and audit logging
- ACID transaction support

Core Tables: Users, Students, Courses, Enrollments, Employees, Payments, Campaigns, Assets

4.3 API Integration

Priority: High

Functional Requirements (REQ-API-01 to REQ-API-07):

- RESTful API architecture
- JWT token authentication
- Comprehensive API endpoints for all modules
- OpenAPI/Swagger documentation
- API versioning (/api/v1/endpoint)
- Rate limiting (1000 requests/hour/user)
- Standardized JSON error responses

Sample Endpoints:

- `POST /api/v1/auth/login`
- `GET /api/v1/courses`
- `POST /api/v1/payments`
- `GET /api/v1/employees`

4.4 Academic Module

Priority: High

REQ-ACD-01: Course Management

- Create, modify, delete courses with code, name, credits, prerequisites

REQ-ACD-02: Student Enrollment

- Enroll students in courses with validation and waitlist support

REQ-ACD-03: Grade Management

- Record and modify grades with approval workflow

REQ-ACD-04: Attendance Tracking

- Mark and track attendance with automated reports

REQ-ACD-05: Examination Management

- Schedule exams, manage seating, publish results

REQ-ACD-06: Academic Dashboard

- Display courses, enrollment stats, grade distribution, attendance rates
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4.5 Marketing & Finance Module

Priority: High

4.5.1 Finance Management

REQ-FIN-01: Fee Management

- Define fee structures, generate invoices, track payment status

REQ-FIN-02: Payment Processing

- Accept multiple payment methods with payment gateway integration

REQ-FIN-03: Expense Management

- Record expenses with approval workflow and budget tracking

REQ-FIN-04: Financial Reporting

- Generate revenue, expense, balance sheet, and cash flow reports

4.5.2 Marketing Management

REQ-MKT-01: Campaign Management

- Create and track marketing campaigns with budget allocation

REQ-MKT-02: Lead Management

- Capture, track, and convert leads with source attribution

REQ-MKT-03: ROI Calculation

- Calculate campaign ROI: $(\text{Revenue} - \text{Cost}) / \text{Cost} \times 100$

REQ-MKT-04: Marketing Dashboard

- Display active campaigns, lead conversion, ROI metrics
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4.6 Administration & Human Resources Module

Priority: High

4.6.1 Employee Management

REQ-HR-01: Employee Records

- Maintain comprehensive employee profiles with document management

REQ-HR-02: Attendance Management

- Track daily attendance with automated reports

REQ-HR-03: Leave Management

- Submit, approve, and track leave requests with balance management

REQ-HR-04: Payroll Processing

- Calculate salaries with deductions and generate payslips

4.6.2 Asset Management

REQ-AST-01: Asset Registration

- Maintain asset inventory with purchase details

REQ-AST-02: Asset Allocation

- Track asset assignments to employees
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5. Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** Web pages $\leq 3s$, API responses $\leq 2s$, database queries $\leq 1s$
- **Throughput:** Support 500 concurrent users, 10,000 transactions/hour
- **Scalability:** Handle 100% user growth over 2 years
- **Capacity:** Store 10+ years historical data, 10,000+ students, 1,000+ employees

5.2 Safety Requirements

- Automated daily backups stored in separate location
- Recovery Time Objective (RTO): 4 hours
- Recovery Point Objective (RPO): 24 hours
- Input validation and referential integrity enforcement
- Graceful error handling with admin notifications

5.3 Security Requirements

- Strong password policy with bcrypt/SHA-256 hashing
- Role-based access control with audit logging
- HTTPS/TLS 1.2+ for all data transmission
- Sensitive data encrypted at rest
- Session timeout after 30 minutes inactivity
- SQL injection, XSS, and CSRF prevention
- Data protection regulation compliance

5.4 Software Quality Attributes

- **Reliability:** 99.5% uptime, MTBF 720 hours, MTTR 2 hours
- **Availability:** 24/7 except planned maintenance (≤ 4 hours/month)
- **Maintainability:** Modular architecture, well-documented code
- **Usability:** Intuitive interface, helpful error messages, context-sensitive help
- **Testability:** 80% minimum test coverage

5.5 Business Rules

- **Academic:** 75% minimum attendance for exam eligibility
- **Financial:** Late payment fees apply after due date

- **HR:** Leave requests 3 days in advance, 6-month probation period
 - **Asset:** Assets returned upon employee separation
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6. System Models

6.1 Use Case Diagrams

6.1.1 Academic Module

[Use Case Diagram illustration for Academic Module]

6.1.2 Finance and Marketing Module

[Use Case Diagram illustration for Finance and Marketing Module]

6.1.3 HR and Administration Module

[Use Case Diagram illustration for HR and Administration Module]

6.2 Class Diagram

[Class Diagram showing system architecture with classes, attributes, methods, and relationships]

6.3 Sequence Diagrams

6.3.1 Course Registration Sequence Diagram

[Sequence Diagram illustration]

6.3.2 Payment Processing Sequence Diagram

[Sequence Diagram illustration]

6.3.3 Leave Request & Approval Sequence Diagram

[Sequence Diagram illustration]

Appendix C: Additional Details

1. Third-Party Integrations

- Specific payment gateway selection
- Email service provider selection
- SMS service provider (if required)
- Biometric device specifications

2. Deployment Environment

- Cloud provider selection (AWS, Azure, Google Cloud)
- Server specifications and configuration
- Domain name and SSL certificate provider

3. Reporting Tools

- Report generation library/tool
- Business intelligence tool integration (optional)

4. Mobile Application

- Decision on native vs hybrid mobile app
- Mobile app features and priority

5. Advanced Features (Future Enhancements)

- AI-powered analytics and predictions
- Chatbot for student/employee queries
- Mobile push notifications
- Advanced scheduling algorithms
- Integration with learning management systems

6. Testing Strategy

- Testing tools and frameworks
- Performance testing benchmarks
- User acceptance testing protocols

7. Training Requirements

- User training program details
- Documentation format and delivery method
- Support model (help desk, documentation, videos)

8. Data Migration Details

- Legacy system data format
 - Migration timeline and approach
 - Data cleansing requirements
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